Deborah Plana, MD, PhD

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EDUCATION AND TRAINING

Massachusetts Institute of Technology ▪ Cambridge, MA *Jun. 2013-Jun. 2017*
Bachelor of Science in Biological Engineering *GPA 4.9/5.0*
Minor in Statistics and Data Science

Harvard Medical School/MIT Medical Scientist Training Program ▪ Boston, MA *Jun. 2017-May 2024*
Harvard PhD in Systems Biology, with the dissertation: "Clinical Trial Data Science to Advance Precision Oncology"
Harvard-MIT MD from the Health Sciences and Technology Program, *magna cum laude*

Massachusetts General Hospital Anesthesia/Research Residency ▪ Boston, MA *Jun. 2024-current*
Pathway to Research Independence with Mentorship and Education (PRIME) track resident

AWARDS

HHMI Exceptional Research Opportunities Program *Jan. 2015*
MIT Ilona Karmel Writing Prize *May 2015*
MIT James E. Cunningham (1957) Memorial Scholar *Dec. 2015*
MGH Summer Research Trainee Program *June-Aug. 2016*
Tau Beta Pi Engineering Honor Society *Sep. 2016*
MIT Reddy Family Undergraduate Research Opportunities Fund *Dec. 2016*
NIH Medical Scientist Training Program *Jun. 2017-May 2024*
HMS Presidential Scholar *Aug. 2017, Aug .2018*
Novartis Quantitative Sciences in Drug Development Best Poster *Aug. 2018*
HMS Harvard Graduate Prize Fellowship *Mar. 2019*
NIH Ruth L. Kirschstein National Research Service Award, F30CA260780 *Jun. 2021-May 2024*
HMS Honors in a Special Field, *magna cum laude* *May 2024*
Sigma Xi Scientific Research Honor Society *Jun. 2024*

RESEARCH, LEADERSHIP, AND WORK EXPERIECE

Undergraduate Researcher ▪ Lauffenburger Lab. ▪ MIT Department of Biological Engineering *Jan. 2015-Jun. 2017*
- Used mixed-effects models of cytokine data from a triculture hydrogel model to investigate the role of different cell types in endometriosis.
- Analyzed RNA-Seq and microarray data from melanoma patient tumor tissue and exosome samples to predict patients' response to immune-checkpoint inhibitors and targeted therapies.

Graduate Researcher ▪ Sorger and Palmer Lab. ▪ HMS Lab. of Systems Pharmacology *Jan. 2018-Mar. 2024*
- Invented a statistical tool to identify treatment-sensitive tumor subgroups in early-phase clinical trials of new therapies.
- Discovered that a parametric distribution (Weibull) describes patient survival data and can estimate drug efficacy with smaller sample sizes as compared to traditional approaches.
- Established a published repository of individual participant survival data from over 150 clinical trials: cancertrials.io.
- Identified biomarkers of drug response in ovarian cancer and lymphoma using patient-derived xenograft (PDX) data.

Co-Founder, Coordinator ▪ Greater Boston Pandemic Fabrication Team ▪ Harvard-MIT *Mar. 2020-Jan. 2022*
- Coordinated team of ~100 scientists, clinicians, and engineers in responding to medical supply shortages during the COVID-19 pandemic, with resulting products used in Boston-area hospitals.
- Guided preparation and publication of multiple open-source research articles on the creation, reuse, and clinical testing of personal protective equipment during a crisis, available at panfab.org.

Research Fellow ▪ Longitude Capital *Aug. 2020-Feb. 2021*
- Sourced and evaluated early-stage companies for biotechnology investment team.
- Investigated therapeutic areas of interest and created presentation materials to summarize research findings.

Technology Venture Fellow ▪ Engine Ventures *Jan. 2024-May 2024*
- Developed landscape of scientific research and company activity addressing unmet needs in fertility.
- Performed technical due diligence for research, analytics, and investment support.

Postdoc. Researcher ▪ Cui Systems Immunology Research Lab. ▪ MGB Dept. of Anesthesiology *Aug. 2025- current*
- Developed mouse experimental system to test the effects of commonly used anesthetics on immune modulation.
- Analyzed single-cell RNA sequencing data to understand how gene expression is modulated by age and drug exposure.

TEACHING

Undergraduate Teaching Assistant ▪ Fields, Forces and Flows (2.793/6.023/20.330) ▪ MIT *Jan. 2017-May 2017*
- Taught weekly recitations and office hours, organized review sessions, graded exams, and moderated Piazza forum.
- Course topics taught included electric fields, fluid flows, transport phenomena and applications to biological systems.
- Received 7.0/7.0 instructor rating.

Graduate Teaching Assistant ▪ Principles and Practices of Drug Development ▪ Harvard-MIT *Aug. 2020-Dec. 2020*
- Created and presented course materials on pharmacology fundamentals, preclinical drug discovery, clinical trials, drug manufacturing, FDA regulation, and financing of new therapeutics for undergraduate and graduate students.

TECHNICAL AND LANGUAGE SKILLS

Data analysis: Mathematica, R, MATLAB, Python
Languages: Native/Bilingual proficiency in English and Spanish. Certified to provide clinical care in Spanish by ALTA language services.

PROFESSIONAL ACTIVITIES

Editorial Fellow: 2026 *Anesthesia & Analgesia* Editorial Fellowship
Ad Hoc Referee: *The Lancet Digital Health, npj Digital Medicine, Scientific Reports, BMJ Open, Frontiers in Medicine, BMC Medical Research Methodology, Interventional Pain Medicine*
Committee and Society Member: Medical School Admissions Committee for Harvard-MIT HST (Jul. 2019-Mar. 2021); Harvard Center for Cancer Systems Pharmacology Executive Committee (May 2020- Mar. 2022)**;** American Society of Anesthesiologists (current); Early Stage Anesthesiology Scholars (current)
Poster Judge: HMS 22nd Annual New England Science Symposium (Apr. 2023)
Research Supervisor: Ivonne Zhou (undergraduate, UNC Chapel Hill), David Huang (undergraduate, Harvard College), William Sorger (undergraduate, Brandeis University), and Harris Davis (undergraduate, UNC Chapel Hill)
Mentor: MIT Office of Academic Community, Empowerment, and Success (OACES) Career Mentor
Co-founder and Contributor: Emergent Properties Blog

PEER-REVIEWED PUBLICATIONS

*Denotes equal contributions

1. Schrier, S. B., Hill, A. S., **Plana, D.**, & Lauffenburger, D. A. (2016). *Synergistic Communication between CD4+ T Cells and Monocytes Impacts the Cytokine Environment*. Scientific Reports, 6(1), 34942.
2. Koch, R., Christie, A. L., Crombie, J. L., Palmer, A. C., **Plana, D.**, Shigemori, K., Morrow, S. N., Van Scoyk, A., Wu, W., Brem, E. A., Secrist, J. P., Drew, L., Schuller, A. G., Cidado, J., Letai, A., & Weinstock, D. M. (2019). *Biomarker-driven strategy for MCL1 inhibition in T-cell lymphomas*. Blood, 133(6), 566–575.
3. Shi, A., Kasumova, G. G., Michaud, W. A., Cintolo-Gonzalez, J., Díaz-Martínez, M., Ohmura, J., Mehta, A., Chien, I., Frederick, D. T., Cohen, S., **Plana, D.**, Johnson, D., Flaherty, K. T., Sullivan, R. J., Kellis, M., & Boland, G. M. (2020). *Plasma-derived extracellular vesicle analysis and deconvolution enable prediction and tracking of melanoma checkpoint blockade outcome*. Science Advances, 6(46), eabb3461.
4. Tyan, K., Levin, A., Avalos-Pacheco, A., **Plana, D.**, Rand, E. A., Yang, H., Maliszewski, L. E., Chylek, L. A., Atta, L., Tye, M. A., Carmack, M. M., Oglesby, N. S., Burgin, S., Yu, S. H., LeBoeuf, N. R., & Kemp, J. M. (2020). *Considerations for the Selection and Use of Disinfectants Against SARS-CoV-2 in a Health Care Setting*. Open Forum Infectious Diseases, 7(ofaa396).
5. Mostaghimi, A.*, Antonini, M.-J.*, **Plana, D.***, Anderson, P. D., Beller, B., Boyer, E. W., Fannin, A., Freake, J., Oakley, R., Sinha, M. S., Smith, L., Van, C., Yang, H., Sorger, P. K., LeBoeuf, N. R., & Yu, S. H. (2020). *Regulatory*

and Safety Considerations in Deploying a Locally Fabricated, Reusable Face Shield in a Hospital Responding to the COVID-19 Pandemic. Med, 1(1), 139-151.e4.

6. Palmer, A. C.*, **Plana, D.***, & Sorger, P. K. (2020). *Comparing the Efficacy of Cancer Therapies between Subgroups in Basket Trials*. Cell Systems, 11(5), 449-460.e2.
7. Palmer, A. C.*, **Plana, D.***, Gao, H.*, Korn, J. M., Yang, G., Green, J., Zhang, X., Velazquez, R., McLaughlin, M. E., Ruddy, D. A., Kowal, C., Muszynski, J., Bullock, C., Rivera, S., Rakiec, D. P., Elliott, G., Fordjour, P., … Sorger, P. K. (2020). *A Proof of Concept for Biomarker-Guided Targeted Therapy against Ovarian Cancer Based on Patient-Derived Tumor Xenografts*. Cancer Research, 80(19), 4278–4287.
8. Kassamali, B., Yu, Z., Davis, C., Carmack, M., Bui, A.-T. N., Said, J. T., **Plana, D.**, Yang, H., Sorger, P., LeBoeuf, N. R., & LaChance, A. H. (2021). *Conversion of Existing UVB Phototherapy Units to UVC Germicidal Chambers for N95 Decontamination: Lessons Learned*. Photobiomodulation, Photomedicine, and Laser Surgery, 39(2), 83–85
9. McAvoy, M.*, Bui, A.-T. N.*, Hansen, C.*, **Plana, D.***, Said, J. T., Yu, Z., Yang, H., Freake, J., Van, C., Krikorian, D., Cramer, A., Smith, L., Jiang, L., Lee, K. J., Li, S. J., Beller, B., Huggins, K., Short, M. P., Yu, S. H., … LeBoeuf, N. R. (2021). *3D Printed frames to enable reuse and improve the fit of N95 and KN95 respirators.* BMC Biomedical Engineering, 3(1), 10.
10. Kothakonda, A.*, Atta, L.*, **Plana, D**.*, Ward, F. *, Davis, C., Cramer, A., Moran, R., Freake, J., Tian, E., Mazor, O., Gorelik, P., Van, C., Hansen, C., Yang, H., Li, Y., Sinha, M. S., Li, J., Yu, S. H., LeBoeuf, N. R., & Sorger, P. K. (2021*). De Novo Powered Air-Purifying Respirator Design and Fabrication for Pandemic Response.* Frontiers in Bioengineering and Biotechnology, 9, 725.
11. Antonini, M.-J.*, **Plana, D.***, Srinivasan, S.*, Atta, L., Achanta, A., Yang, H., Cramer, A. K., Freake, J., Sinha, M. S., Yu, S. H., LeBoeuf, N. R., Linville-Engler, B., & Sorger, P. K. (2021). *A Crisis-Responsive Framework for Medical Device Development Applied to the COVID-19 Pandemic*. Frontiers in Digital Health, 3, 617106.
12. Cramer, A. K.*, **Plana, D.***, Yang, H.*, Carmack, M. M., Tian, E., Sinha, M. S., Krikorian, D., Turner, D., Mo, J., Li, J., Gupta, R., Manning, H., Bourgeois, F. T., Yu, S. H., Sorger, P. K., & LeBoeuf, N. R. (2021). *Analysis of SteraMist ionized hydrogen peroxide technology in the sterilization of N95 respirators and other PPE*. Scientific Reports, 11(1), 2051.
13. **Plana, D.***, Tian, E.*, Cramer, A. K.*, Yang, H.*, Carmack, M. M., Sinha, M. S., Bourgeois, F. T., Yu, S. H., Masse, P., Boyer, J., Kim, M., Mo, J., LeBoeuf, N. R., Li, J., & Sorger, P. K. (2021*). Assessing the filtration efficiency and regulatory status of N95s and nontraditional filtering face-piece respirators available during the COVID-19 pandemic*. BMC Infectious Diseases, 21(1), 712.
14. Ye, Z., Qian, J. M., Hosny, A., Zeleznik, R., **Plana, D**., Likitlersuang, J., Zhang, Z., Mak, R. H., Aerts, H. J. W. L., & Kann, B. H. (2022). *Deep Learning–based Detection of Intravenous Contrast Enhancement on CT Scans.* Radiology: Artificial Intelligence, 4(3), e210285.
15. Peterson, J. S., **Plana, D.**, Bitterman, D. S., Johnson, S. B., Aerts, H. J. W. L., & Kann, B. H. (2022). *Growth in eligibility criteria content and failure to accrue among National Cancer Institute (NCI)-affiliated clinical trials*. Cancer Medicine, 12(4), 4715–4724.
16. **Plana, D.**, Shung, D. L., Grimshaw, A. A., Saraf, A., Sung, J. J. Y., & Kann, B. H. (2022). *Randomized Clinical Trials of Machine Learning Interventions in Health Care: A Systematic Review.* JAMA Network Open, 5(9), e2233946.
17. **Plana, D.***, Palmer, A. C.*, & Sorger, P. K. (2022*). Independent Drug Action in Combination Therapy: Implications for Precision Oncology*. Cancer Discovery, 12(3), 606–624.
18. **Plana, D.**, Fell, G., Alexander, B. M., Palmer, A. C., & Sorger, P. K. (2022). *Cancer patient survival can be parametrized to improve trial precision and reveal time-dependent therapeutic effects.* Nature Communications, 13(1), 873.
19. Kazmierski, M., Welch, M., Kim, S., McIntosh, C., Rey-McIntyre, K., Huang, S. H., Patel, T., Tadic, T., Milosevic, M., Liu, F.-F., Ryczkowski, A., Kazmierska, J., Ye, Z., **Plana, D**., Aerts, H. J., Kann, B. H., Bratman, S. V., Hope, A. J., & Haibe-Kains, B. (2023). *Multi-institutional prognostic modelling in head and neck cancer: evaluating impact and generalizability of deep learning and radiomics*. Cancer Research Communications, CRC-22-0152
20. Hwangbo, H., Patterson, S. C., Dai, A., **Plana, D.**, & Palmer, A. C. (2023). *Additivity predicts the efficacy of most approved combination therapies for advanced cancer*. Nature Cancer, 1–12.
21. Zhou, I. *, **Plana, D.** *, & Palmer, A. C. (2024). *Tumor-specific activity of precision medicines in the NCI-MATCH trial*. Clinical Cancer Research, 30 (4): 786–792. *These authors contributed equally.
22. Pomeroy, A., Sworder, B., **Plana, D.**, Cao, Y., Alizadeh, A., & Palmer, A. C. (2025). *A pharmacokinetic/pharmacodynamic model predicts tumor debulking improves CAR T-cell efficacy in Large B-Cell Lymphoma.* Blood Cancer Discovery, BCD-25-0138.

ORAL PRESENTATIONS

1. **Plana, D.** *Using blood-based assays and RNA data to predict melanoma patient response to therapy*. Summer Research Trainee Program at the Massachusetts General Hospital. Boston, MA. June 2016.
2. **Plana, D.** *Parametric fitting of clinical trial data in oncology*. Harvard Institute for Therapeutic Sciences Symposium. Boston, MA. October 2019.
3. **Plana, D.** *Of mice and (small numbers of) men: Improving estimates of drug efficacy in oncology.* Harvard Systems Biology Department Pizza Talk. Boston, MA. October 2019.
4. **Plana, D.** *Parametric fitting of over 100,000 individual patient events from clinical trials in oncology*. Harvard-MIT Center for Regulatory Science Doctoral Student Symposium. Boston, MA. February 2020.
5. **Plana, D.** *Introduction to PanFab: Mobilizing a volunteer response to COVID-19* (Speaker); *Resiliency and Trust: Local, regional, and national response to PPE shortages* (Panelist). Harvard-MIT Center for Regulatory Science Pandemic Response: Resilience and Recovery in the Era of COVID-19. Virtual. June 2020.
6. **Plana, D.** *Making Precise Estimates of Drug Efficacy from Small Amounts of Patient Data.* Tumor Therapeutics and Resistance Session; Junior Investigator Meeting for the NCI Cancer Systems Biology Consortium, Physical Sciences-Oncology Network, and the Big Data Scientist Training Enhancement Program. Virtual. August 2020.
7. **Plana, D.**, Palmer, A.C., Sorger, P.K. *Advancing the design and interpretation of cancer clinical trials through systematic data mining and analysis.* HMS Ludwig Cancer Center. Virtual. June 2021.
8. **Plana, D.** *Biological Engineering as a Foundation for a Career in Medical Research.* 20.001: MIT Introduction to Professional Success and Leadership in Biological Engineering. Cambridge, MA. November 2022.
9. **Plana, D.** *Highlights from the Bigelow: Residency Intake.* Massachusetts General Hospital Department of Medicine Noon Conference. Boston, MA. April 2023.
10. **Plana, D.** *Where are they now? Alumni Panel*. MIT ACCESS Program. Boston, MA. October 2025.
11. **Plana, D.** *Ketamine Polypharmacology.* Massachusetts General Hospital Department of Anesthesia, Critical Care and Pain Medicine Journal Club. Boston, MA. November 2025.
12. **Plana D**., Wang L., Cui A. *Anesthetic immunomodulation at single-cell resolution.* E-Poster with Moderated Oral Presentation at the New England Anesthesia Resident Conference (NEAR). Boston, MA. September 2026

POSTER PRESENTATIONS

1. **Plana, D.**, Cocchi, M., Sundar, E., Pleumann, A., Grzybinski, M., Shaefi, S., Brogna, M. J., Sisti, L., Butler, K. L. Small, A., Cain, C., Muret-Wagstaff, S., Foley, J., Talmor, D., Ma, H., Whyte, R., Kleinman, G., Simon, B. A. *Early Identification and Care of the Critically Ill Post-Operative Patient*. Beth Israel Deaconess Medical Center Silverman Symposium. Boston, MA. April 2015.
2. **Plana, D.**, Hill, A., Cook, C., Griffith, L., Lauffenburger, D. *Creating a 3-D hydrogel model of the human endometrium and its interactions with immunological factors*. New England Science Symposium. Cambridge, MA. April 2016.
3. **Plana, D.**, Wang, S., Cepko, C. *Characterization of retinal cell subpopulations expressing distinct Otx2 enhancers.* Howard Hughes Medical Institute Exceptional Research Opportunities Program Meeting. Chevy Chase, MD. May 2016.
4. Cintolo-Gonzalez, J., Michaud, W., Cohen, S., **Plana, D.**, Panka, D., Sullivan, R., Boland, G. *Use of Circulating Microvesicles, Exosomes, as a Biomarker to Track Disease Burden in Melanoma*. Massachusetts General Hospital Clinical Research Day. Boston, MA. October 2016.
5. **Plana, D.**, Hill, A., Cook, C., Griffith, L., Lauffenburger, D. *Characterizing a 3-D hydrogel model of the human endometrium and its interactions with immunological factors using mixed effects modeling*. Biomedical Engineering Society Conference. Minneapolis, MN. October 2016.
6. **Plana, D.**, Palmer, A. C., Sorger, P.K**.** *Predicting patient response to cancer combination therapies from clinical trial data*. Novartis-Academia Conference on Quantitative Sciences in Drug Development. Cambridge, MA. August 2018.
7. **Plana, D.**, Palmer, A. C., Sorger, P.K**.** *Making Precise Estimates of Drug Efficacy from Small Amounts of Patient Data*. NCI Center for Cancer Systems Pharmacology Annual Investigators Meeting. Virtual. September 2020.
8. **Plana, D.**, Fell, G., Alexander, B. M., Palmer, A. C., & Sorger, P. K. *Cancer patient survival can be accurately parameterized, revealing time-dependent therapeutic effects and doubling the precision of small trials.* Keystone Symposia, Precision Oncology: Translating Discovery to the Clinic. Virtual. June 2021.
9. **Plana, D.**, Fell, G., Alexander, B. M., Palmer, A. C., & Sorger, P. K. *Re-analysis and sharing of clinical trial data to advance precision oncology.* 2021 Cancer Systems Biology Consortium Annual Investigator Meeting. Virtual. September 2021.
10. **Plana, D.**, Guntaka, P. K., Qian, J. M., Zhou, P., Hosny, A., Margalit, D. N., Schoenfeld, J. D., Tishler, R. B., Haddad, R. I., Uppaluri, R., Haibe-kains, B., Aerts, H., Kann, B. H. *Deep Learning and Harmonization of Multi-Institutional Data for Automated Gross Tumor and Nodal Segmentation for Oropharyngeal Cancer*. American Society for Radiation Oncology Annual Meeting. Virtual. October 2021.

11. **Plana, D.**, Fell, G., Alexander, B. M., Palmer, A. C., & Sorger, P. K. *Cancer patient survival can be accurately parameterized, improving trial precision and revealing time-dependent therapeutic effects.* Pacific Symposium on Biocomputing (PSB). Big Island, HI. January 2022.

DIGITAL PUBLICATIONS

1. **Plana, D**., Arfè, A., & Sinha, M. S. (2020). *Re-Envisioning Clinical Trials During The COVID-19 Pandemic.* Health Affairs Blog.
2. **Plana, D.**, Arfe, A., & Sinha, M. S. (2020). *FDA-2020-D-1106-0139, Re: FDA Guidance on Conduct of Clinical Trials of Medical Products during COVID-19 Pandemic; Guidance for Industry, Investigators, and Institutional Review Boards*. Regulations.gov.
3. **Plana, D.** (2022). *Data science can help us run better cancer clinical trials*. Mathematical Oncology Blog.
4. Desai, V., Duggan, J., Glowala, J., Gupta, M., Kim, K., Liu, G., McDaniel, K., Nabel, K., Nagireddy, H., **Plana**, **D.**, Rencsok, E.; Verheyen, C., Zhong, L. *Module 2: Epidemiology Principles*. (2020). HMS Medical Student COVID-19 Curriculum.
5. Gupta, A., **Plana, D**., Antonini, M.J., Cramer, A., McAvoy, M., Atta, L. *HST students share their reflections on responding to Covid-19*. (2020). Harvard-MIT Health Sciences and Technology News.
6. Winter, C. and **Plana, D**. *The six pillars of thoughtful data generation.* (2025). Emergent Properties Blog.